Chapter 5: Operating Systems

An Operating System (OS) is the most critical software in a computer system. It acts as an intermediary between the computer hardware and the user, managing resources and providing a platform for application software to run.

5.1 Introduction

Hardware: Physical components of the computer (Mouse, Keyboard, Hard Disk).

Firmware: Instructions stored in ROM (BIOS) that help start the computer.

Software: Sets of instructions to perform specific tasks.

The Booting Process

Booting is the process of loading the OS from the hard disk into the RAM. Steps include:

- **POST (Power On Self-Test):** Checks if all hardware is functioning.
- **BIOS:** Activates and identifies the boot drive.
- **Bootstrap Loader:** Loads the OS into memory.

5.2 Types of Software

Category	Description	Examples
System Software	Manages hardware and provides a platform for applications.	Windows, Linux, Utility tools.
Application Software	Helps users perform specific tasks.	Web Browsers, Games, Word Processors.

5.3 Functions of an Operating System

- **Process Management:** Managing the CPU's execution of tasks.
- **Memory Management:** Allocating space in RAM for active programs.
- **Device Management:** Controlling input/output devices (Printers, Keyboards).
- **File Management:** Organizing data into files and folders.
- **Security:** Protecting data through passwords and permissions.

5.4 Common Operating Systems

- **Microsoft Windows:** Proprietary; popular for PCs.
- **Apple Mac OS:** Proprietary; exclusive to Apple hardware.
- **Ubuntu:** Free and Open Source (Linux-based).
- **Android:** Open Source; primarily for mobile devices.

5.5 Classification of OS

- **Single User:** Serves one person at a time (e.g., MS-DOS).
- **Multi-User:** Allows multiple users to access the system simultaneously (e.g., Windows Server).
- **Multi-Tasking:** Allows a user to run several programs at once (e.g., Windows 10, Ubuntu).
- **Real-Time:** Processes data without delay (e.g., used in medical devices or air traffic control).

Summary

- The OS is essential for managing computer resources.
- User Interfaces can be CLI (Command Line) or GUI (Graphical User Interface).
- Files are stored in folders within drives to keep data organized.